8.7 Answers: ready for the brugklas

← Unit overview · Practice

For the parent: these are new ideas. Any answer that shows he used the number line, the square picture or the machine is a good sign, even if a number slipped.

Set A: Below zero

1. **−2 °C.** From 3, count down 5: 2, 1, 0, −1, −2.
2. **4.** From −2, count up 6.
3. **−3.** From 5, count down 8, going 3 past zero.
4. **−5, −1, 0, 2.** Further below zero is smaller.
5. **−7.** From −4, count down 3 more.
6. **10 degrees.** 6 up to zero, then 4 more.
7. **−€5.** She spent €5 more than she had.
8. **3 floors.** 2 → 1 → 0 → −1.
9. **−3 °C.** It's nearer to zero, so less cold.
10. **0.** One up from −1.

Set B: Squares and roots

1. **9.** 3 × 3.
2. **25.** 5 × 5.
3. **100.** 10 × 10.
4. **1.** 1 × 1.
5. **4.** 4 × 4 = 16.
6. **7.** 7 × 7 = 49.
7. **10.** 10 × 10 = 100.
8. **6 cm.** 6 × 6 = 36.
9. **144.** 12 × 12 = 120 + 24.
10. **Between 7 and 8.** $7^2 = 49$ is just under 50; $8^2 = 64$ is over.
11. **8.** 2 × 2 = 4, × 2 = 8.

Set C: Number machines and formulas

1. **7.** 3 × 2 = 6, + 1 = 7.
2. **11 and 21.** 5 × 2 + 1; 10 × 2 + 1.
3. **10.** 6 + 4.
4. **7.** Backwards: 15 − 1 = 14, then 14 : 2 = 7. (Backwards means undo each step in reverse order.)
5. **14.** 3 × 4 = 12, + 2.
6. **Rule: × 3, then + 2. In 10 → out 32.** The outputs go up by 3 each time, and 3 × 1 + 2 = 5 fits.
7. **€35.** 10 × 3 = 30, + 5.
8. **$3a = 12$, $a + 7 = 11$, $a^2 = 16$.**
9. **€13.** 50 × €0,10 = €5; €8 + €5.
10. **0.** Anything × 0 is 0.

Set D: Which comes first?

1. **14.** 3 × 4 = 12 first; 2 + 12.
2. **20.** Brackets first: 5; 5 × 4.
3. **7.** 6 : 2 = 3 first; 10 − 3.
4. **2.** Brackets first: 4; 4 : 2.

5. **10.** $3^2 = 9$ first; 9 + 1.
6. **25.** × and : have equal rank, so left to right: 20 : 4 = 5, then 5 × 5.
7. **11.** Left to right: 18 − 4 = 14, then 14 − 3.
8. **25.** Brackets first: 5; 5 × 5.
9. **18.** Power first: $3^2 = 9$; 2 × 9.
10. **75.** 10 × 3 = 30 first; then left to right: 100 − 30 = 70, 70 + 5.
11. **2.** 1 × 1 = 1 first; 1 + 1.

Puzzle corner

1. **4, 9, 16: square numbers!** Adding the next odd number adds an L-shaped strip around a square to make the next square. Draw it with squared paper.
2. **Biggest: (4 + 4) × 4 − 4 = 28. Smallest: (4 + 4) × (4 − 4) = 0.** Without brackets it's 16.
3. **3.** 3 × 3 = 9, 9 − 6 = 3.